

EDU-TRACK

Predict.Improve.Succeed

Student Performance Predicting
System Using Random Forest
Algorithm



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ABSTRACT

The Student Performance Predicting System Using Random Forest Algorithm is a webbased machine learning application developed to predict students' academic performance and provide personalized recommendations for academic improvement. The system analyze multiple factors including study hours, attendance percentage, sleep duration, social media usage, previous academic scores, participation in extracurricular activities, part-time employment status, internet accessibility, and parental education level. Based on these inputs, the system classifies students into performance categories such as High, Medium, and Low using a trained Random Forest Classifier. The application calculates prediction confidence scores, determines academic risk levels, and generates personalized recommendations aimed at improving study habits, learning efficiency, attendance, and overall academic achievement. The system is implemented using Python, Flask, Scikit-Learn, SQLite, HTML, CSS, JavaScript, and Chart.js. It incorporates role-based authentication for Students, Teachers, and Administrators, along with prediction history management, dashboard analytics, graphical visualizations, and performance monitoring features. The proposed system supports data-driven educational decision-making by enabling early identification of students requiring academic assistance, facilitating continuous performance monitoring, and providing actionable insights to enhance student success. By integrating machine learning techniques with web-based analytics, the system offers an effective and scalable solution for modern educational institutions



INTRODUCTION



- Predicting student performance is essential for improving academic success and reducing failure rates.
- Traditional evaluation methods rely on manual analysis and often fail to identify struggling students early.
- With the advancement of Artificial Intelligence and Machine Learning, it is now possible to analyze multiple factors affecting student performance.
- This system uses academic and lifestyle data such as study hours, attendance, sleep patterns, and social media usage.
- The goal is to provide accurate predictions and meaningful insights to support students, teachers, and institutions in making data-driven decisions.

LITERATURE SURVEY:

TOPIC	PUBLISHER & Y.O..P	SUMMARY	LINK
Predicting Students Academic Performance using an Improved Random Forest Classifier	IEEEExplore 2020	This paper introduces an Improved Random Forest Classifier to predict student academic success and identify at-risk students early. By analyzing key performance factors, the proposed method achieves higher accuracy than standard algorithms like Naive Bayes, Bagging, and traditional Random Forest.	https://ieeexplore.ieee.org/document/9167547/
Research on Student Achievement Prediction Method Based on Machine Learning	IEEEExplore 2023	This paper addresses the lack of early-warning systems for failing grades in universities. It proposes a machine learning method to predict students' future grades based on their previous semester test scores. The model achieved a prediction accuracy of 88.78%, helping teachers identify at-risk students and implement targeted teaching strategies.	https://ieeexplore.ieee.org/document/10242700/
Student Performance Prediction Using Machine Learning	International Journal of Creative Research Thoughts (IJCRT) 2025	This study utilizes machine learning to predict academic outcomes by analyzing attendance, past grades, and behavioral data. The researchers found that ensemble models, specifically XGBoost (95.1% accuracy) and Random Forest, are the most effective for early identification of at-risk students. A real-world application showed that early intervention based on these predictions helped 75% of flagged students improve their performance.	https://ijcrt.org/papers/IJCRT2507142.pdf

LITERATURE SURVEY:

TOPIC	PUBLISHER & Y.O..P	SUMMARY	LINK
Prediction of Student Academic Performance in Higher Education Using Machine Learning Techniques	International Journal for Multidisciplinary Research 2025	This study utilizes machine learning algorithms like Random Forest and SVM to predict student performance based on academic records and demographics. The research identifies previous grades and attendance as the most influential factors for success. Results show that ensemble-based models provide the highest accuracy, offering a data-driven framework for early identification of at-risk students.	https://www.ijfmr.com/papers/2025/6/64390.pdf
A Systematic Literature Review of Student' Performance Prediction Using Machine Learning Techniques	MDPI 2021	This paper provides a comprehensive review of research from 2009 to 2021 on using machine learning to predict student performance. It identifies key data sources like academic marks and digital activity logs, noting that ensemble models and transfer learning are highly effective at identifying at-risk students. These predictive tools help institutions improve retention through timely, data-driven interventions.	https://www.mdpi.com/2227-7102/11/9/552
A Machine Learning Approach for Predicting Academic Performance and Risk Levels.	International Journal of Innovative Research in Technology 2026	This paper details an AI-based system that uses a Random Forest Regressor to predict student scores and categorize them into High, Medium, or Low risk levels. By analyzing factors like attendance, study hours, and health, the system provides teachers with an analytics dashboard and students with personalized, AI-generated study plans. Experimental results confirm the system effectively identifies at-risk students in real-time, allowing for early academic intervention.	https://ijirt.org/publishedpaper/IJIRT190556_PAPER.pdf

PROBLEM STATEMENT

Educational institutions often face significant challenges in identifying students who are at risk of poor academic performance at an early stage. Traditional evaluation methods primarily rely on past academic results, which provide a limited view and fail to consider important lifestyle factors such as study habits, sleep patterns, and daily activities. This lack of a comprehensive, data-driven approach makes it difficult for educators to perform accurate analysis and take timely corrective actions. As a result, there is a growing need for an intelligent system that can analyze multiple academic and behavioral factors simultaneously and provide reliable predictions of student performance, enabling early intervention and improved educational outcomes.

DRAWBACKS OF EXISTING SYSTEMS

Data & Contextual Limitations

Synthetic Data Reliance: Many systems are tested on artificially generated data, which may not accurately represent the complexities of real-world classrooms.

Missing Qualitative Factors: Models frequently overlook critical non-academic variables such as emotional health, socio-economic status, and individual learning styles.

Delayed Indicators: Some systems rely heavily on previous semester scores, failing to account for sudden changes in a student's current circumstances.

Technical & Implementation Challenges

The "Black Box" Problem: High-accuracy complex models often lack transparency, making it difficult for educators to understand the reasoning behind a student's risk level.

Scalability Issues: Certain applications are built for offline, local environments, preventing large-scale deployment across multiple institutions.

Inconsistent Accuracy: Prediction reliability varies significantly (ranging from 70% to 95%) depending on which specific data features are selected.

PROPOSED SOLUTION

To address these challenges, the proposed system leverages **Artificial Intelligence and Machine Learning** techniques to predict student performance more accurately and efficiently. The system analyzes both academic and lifestyle factors, such as study hours, attendance, sleep patterns, and daily habits, to provide a holistic evaluation of each student. Based on this analysis, students are classified into performance categories like Low, Medium, and High. Additionally, the system offers interactive dashboards for students, teachers, and administrators, enabling better insights and monitoring. This approach supports early identification of at-risk students and promotes data-driven decision-making to improve overall academic outcomes.

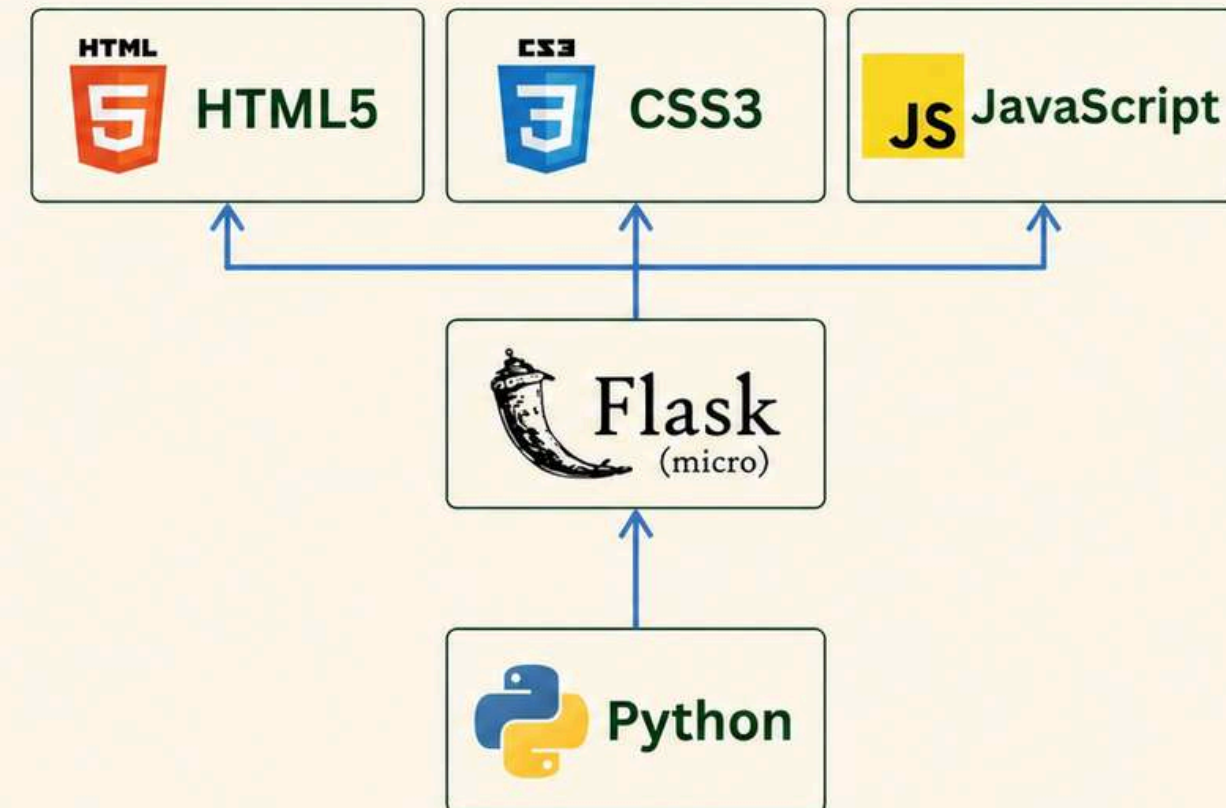


TECH - STACK

Database

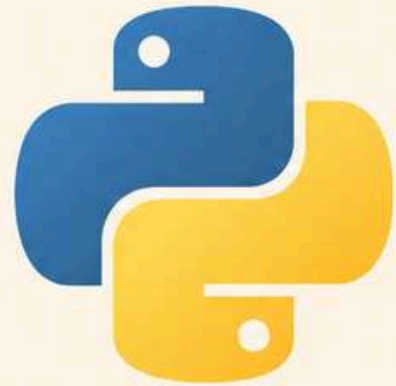


Web Development Tech



TECH - STACK

Machine Learning Tech



PYTHON

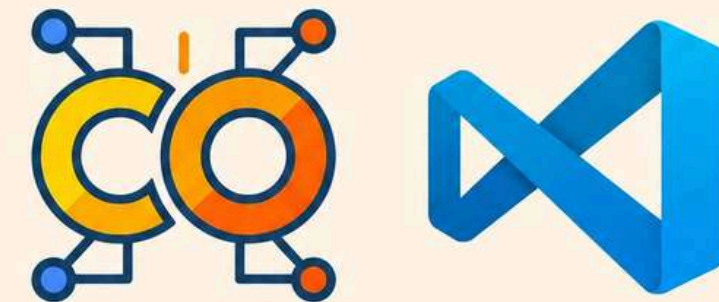


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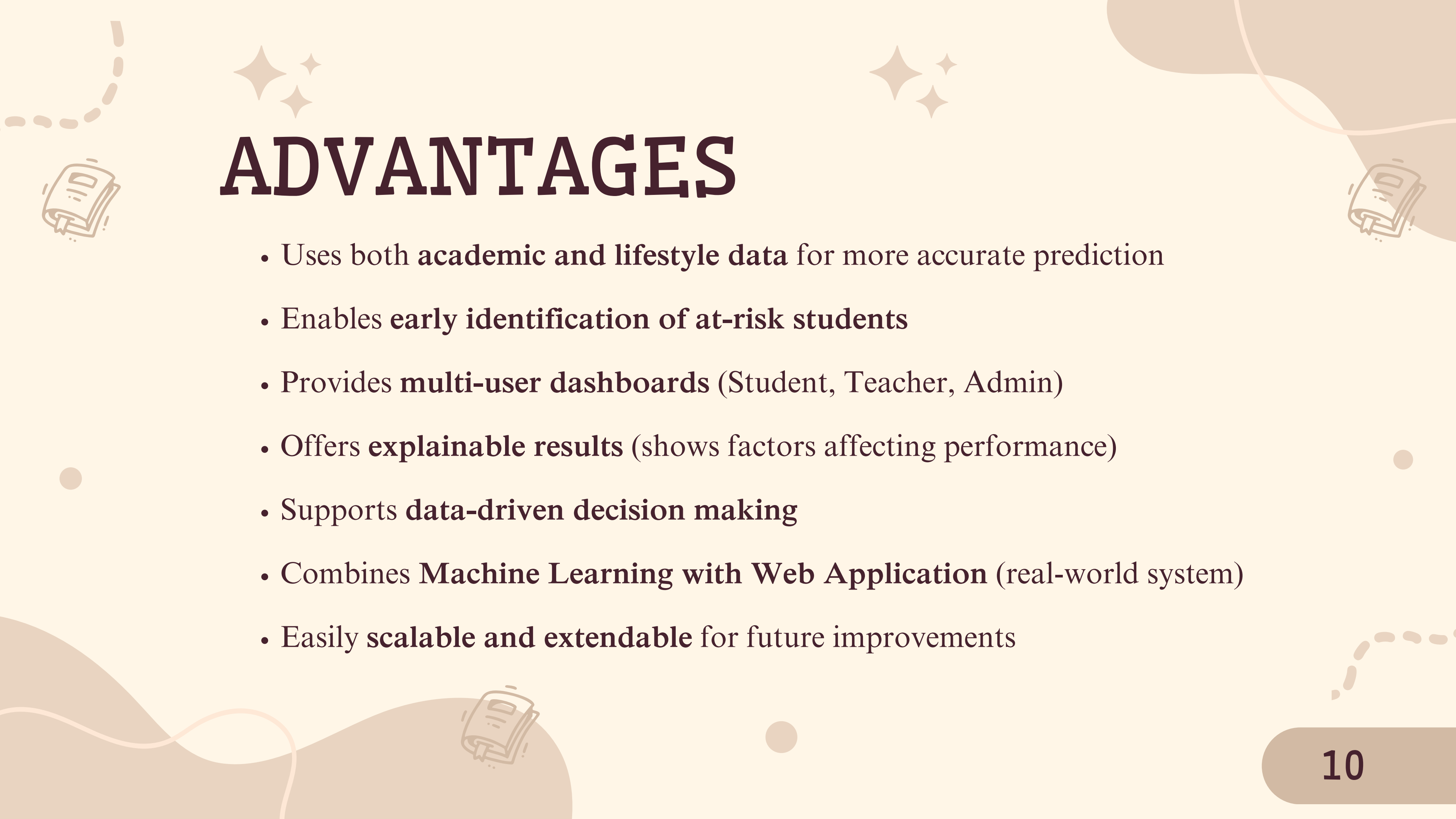


SCIKIT-LEARN

Development Tools Tech



**GOOGLE COLAB
VS CODE**



ADVANTAGES

- Uses both **academic and lifestyle data** for more accurate prediction
- Enables **early identification of at-risk students**
- Provides **multi-user dashboards** (Student, Teacher, Admin)
- Offers **explainable results** (shows factors affecting performance)
- Supports **data-driven decision making**
- Combines **Machine Learning with Web Application** (real-world system)
- Easily **scalable and extendable** for future improvements

✦ ✦ FUTURE SCOPE ✦ ✦

- Integration with real-time student data from schools/colleges
- Development of a mobile application for easier access
- Use of advanced ML models (XGBoost, Deep Learning) for higher accuracy
- Addition of personalized recommendation system for students
- Integration with attendance and LMS systems
- Expansion to multiple institutions for large-scale analysis
- Inclusion of real-time alerts/notifications for at-risk students
- Support for predicting performance across different subjects

CONCLUSION

The Student Performance Predicting System Using Random Forest Algorithm successfully demonstrates how Machine Learning can be used to analyze and predict student performance based on both academic and lifestyle factors. By classifying students into performance categories, the system enables early identification of at-risk students and supports timely intervention. The integration of student, teacher, and admin dashboards provides meaningful insights and promotes data-driven decision-making. Overall, this project highlights the potential of AI in improving educational outcomes and can be further enhanced with real-time data and advanced features in the future.

THANK YOU

